

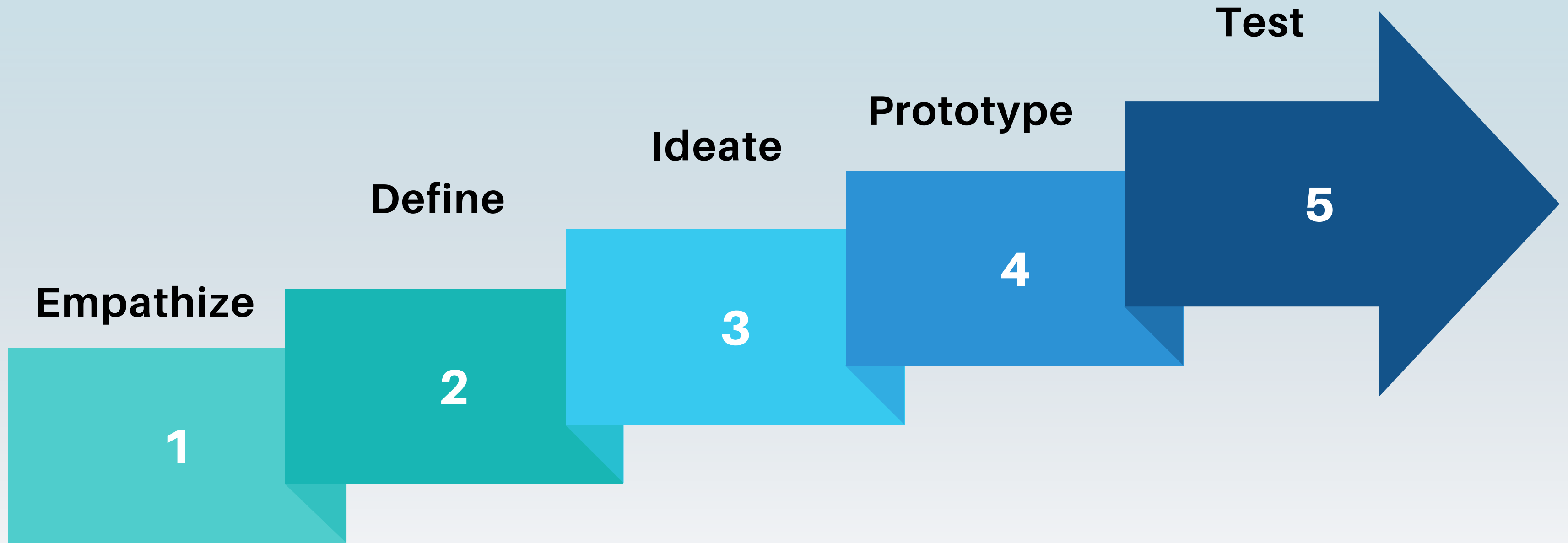
ONLINE CENSUS

DESIGN THINKING

~ BY
NANDINI



5D PROCESS



Problem Statement

The current offline census system is slow, resource-heavy, and prone to human error. Digitalizing the process can reduce cost, increase accuracy, ensure transparency, and make census data reliable and timely.

Main Idea Thoughts:

Computerised
verification

digitalizing
while securing

What is the problem?

- Burden of expenses
- Depletion of resources
- Years long process
- Inaccurate results
- Missing data
- Ill-treatment of enumerators

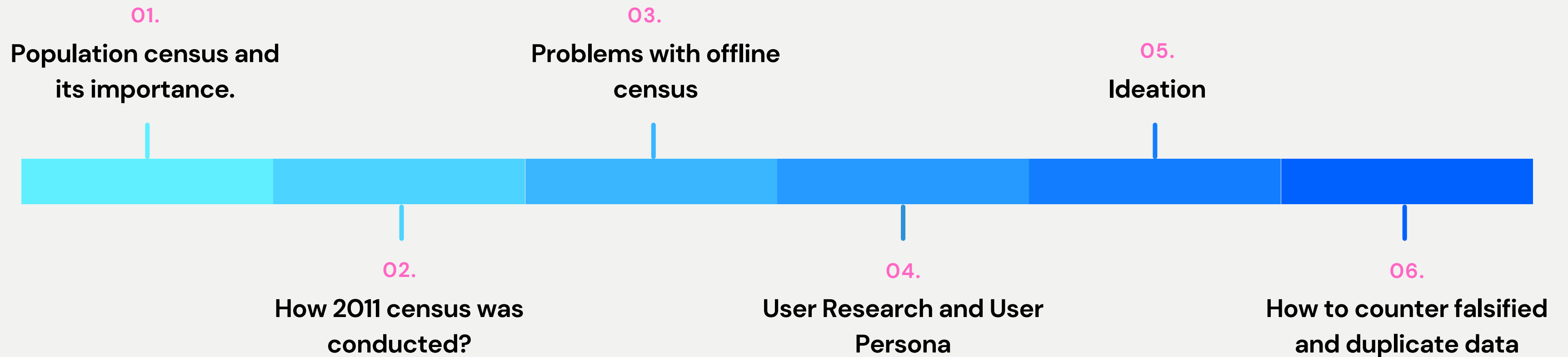
How do we know this is a problem?

- Cost of papers , manual labor and their trainings, etc. during the process
- Increased consumption of food , petrol , paper, etc.
- Long period of time taken in collecting , storing and producing the data
- Manual error, Enumerator - bias , etc.
- Absence of people at their places , voluntarily hiding details
- News and Reviews from govt enumerators

How will we know when we've solved this problem?

- Reduced use of government resources.
- Eliminating and reducing the usage of paper, fuel, manual labor ,etc. used.
- Time reduced to days providing more independence to conduct more frequent census.
- Automated verification
- Time flexibility. No hesitation in sharing data.
- Reduced human involvement

Approach to the problem

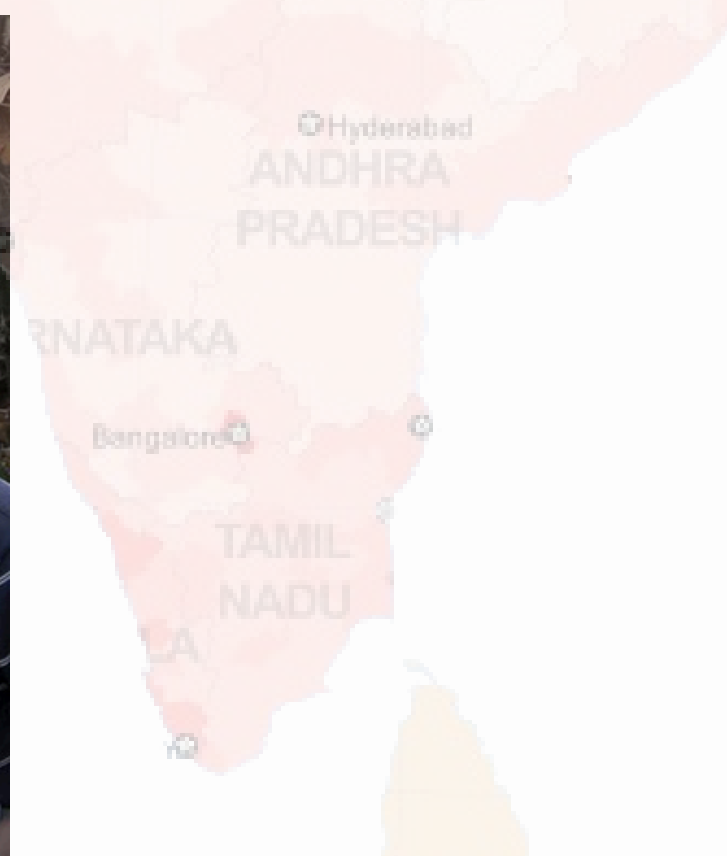


How Might We:

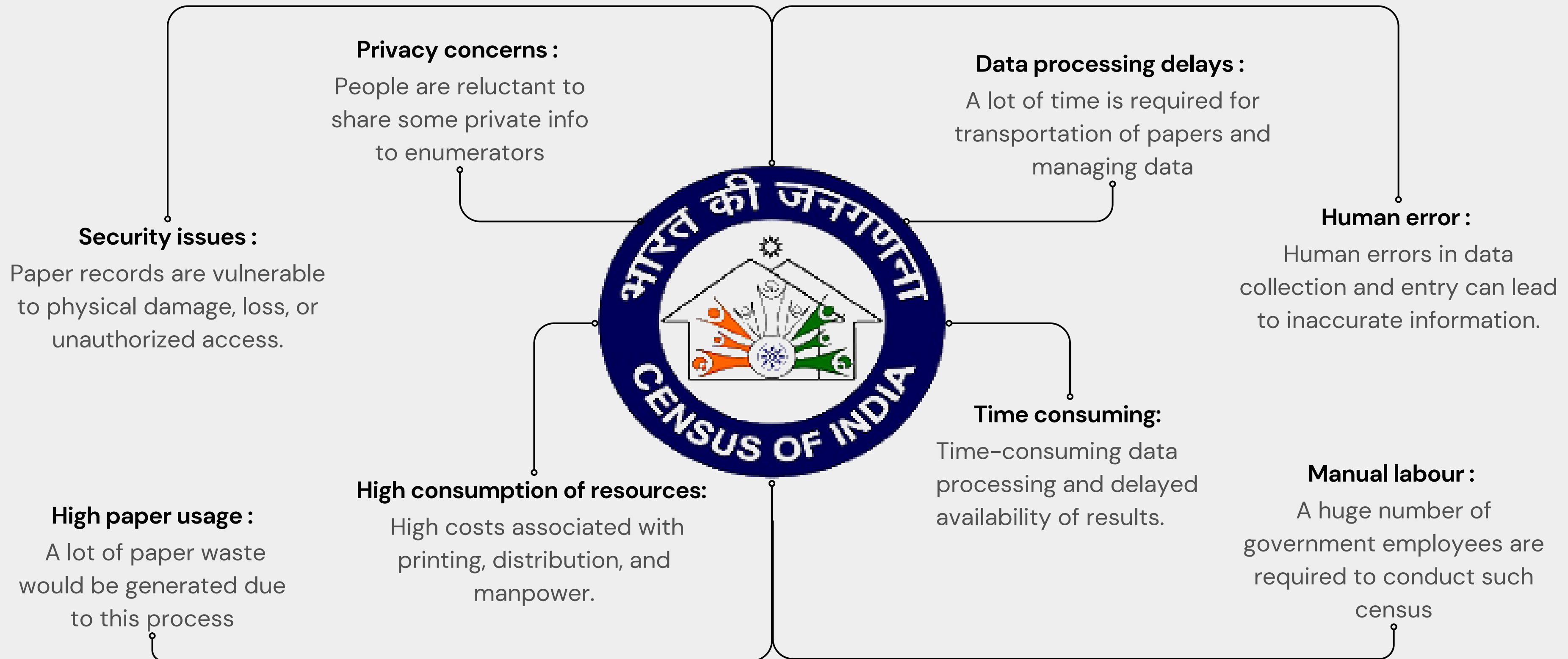
- How might we digitalize census collection to reduce time and cost?
- How might we make data submission safer and more accurate?
- How might we create a system that prevents duplicate and false entries?
- How might we make the process comfortable for citizens of all digital levels?
- How might we simplify the workflow for government enumerators?



INDIAN GOVERNMENT CONDUCTING OFFLINE POPULATION CENSUSES

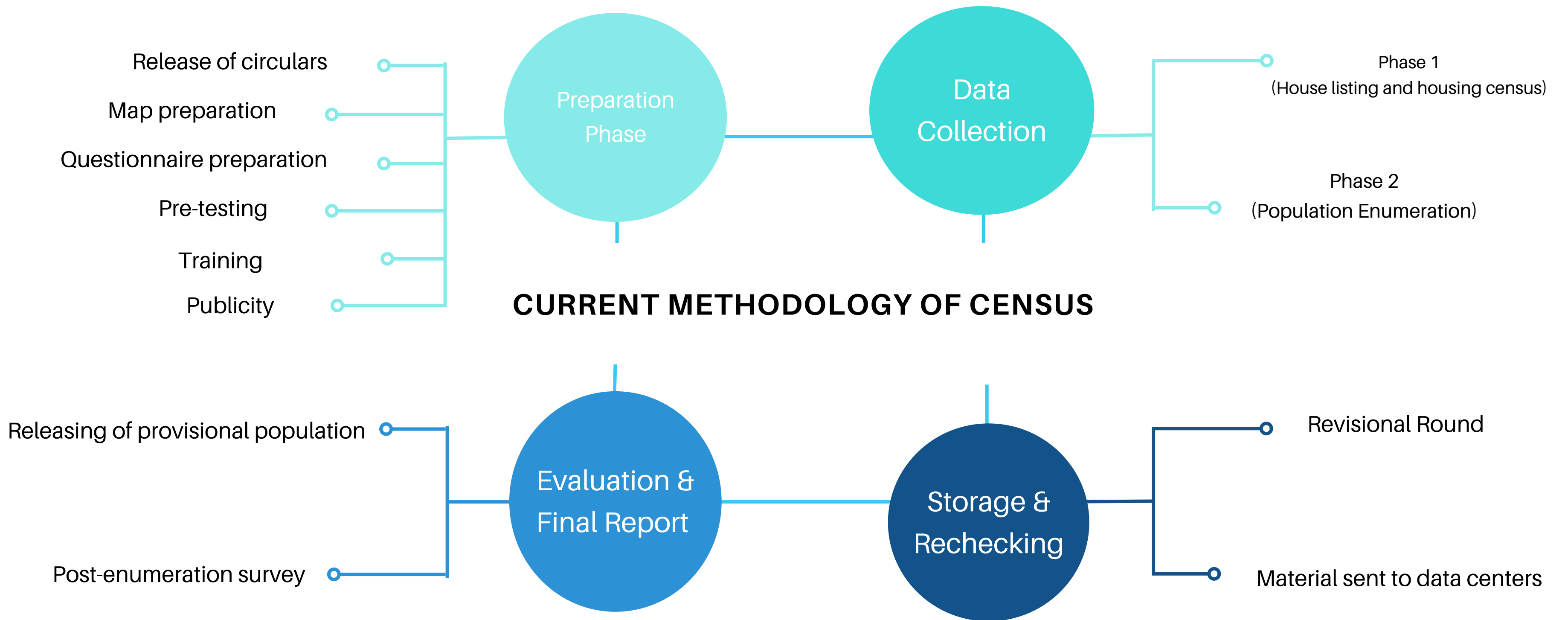


Problems with offline census



Impacts due to census not taking place :

- We are unable to perform census again
- Using old database which was taken in account back in 2011
- Due to not updated database alot resources are being wasted
- Problem and Unfair being used in elections



Analysis



Constructing



information



Producing

ADVANTAGES OF ONLINE CENSUS



**SIMPLIFIED
PROCESS**



**EASY TO
ANALYSE AND
PRODUCE DATA**

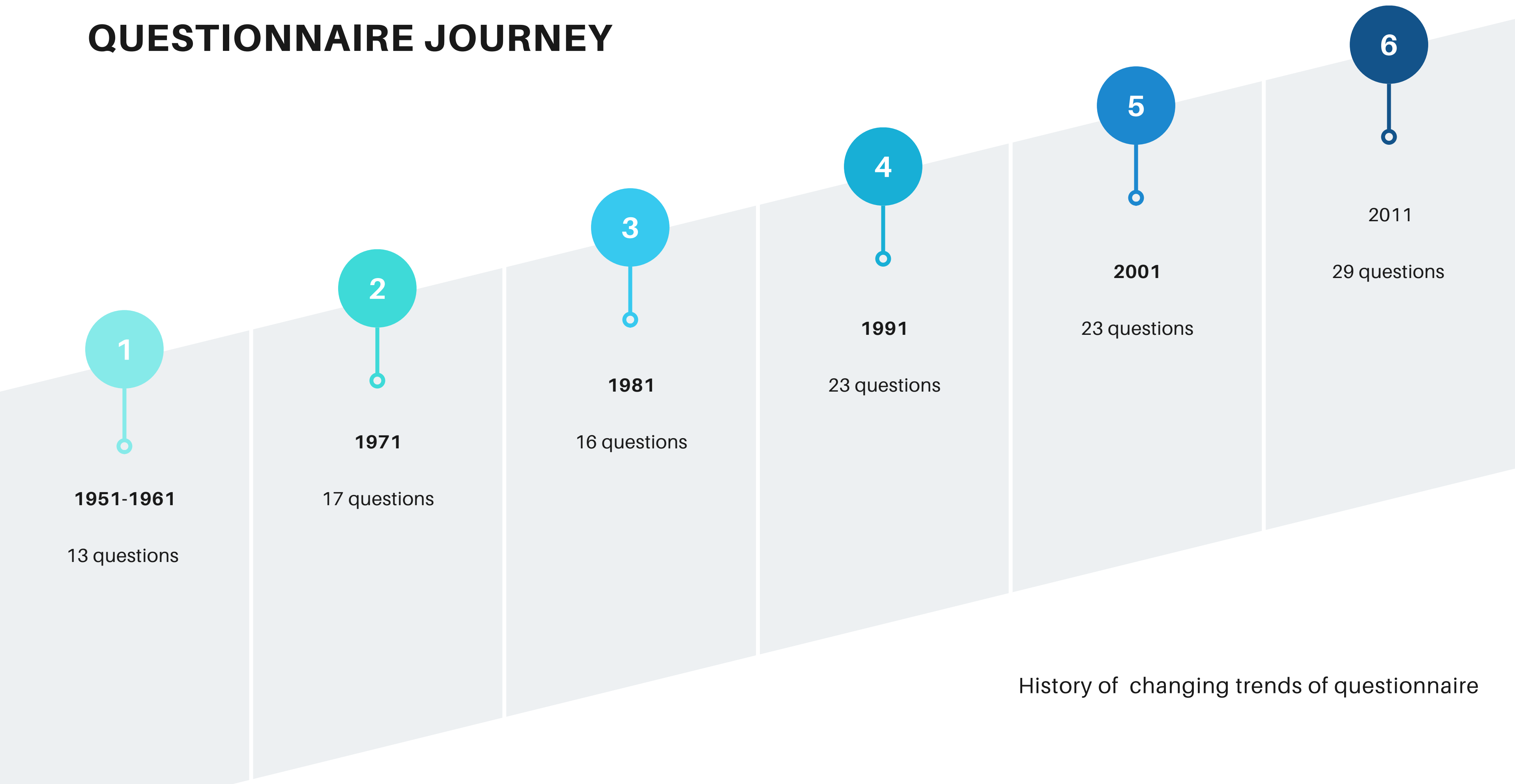


**CAPTURES
PRECISELY**



EASY TO STORE

QUESTIONNAIRE JOURNEY



1951-1961

13 questions

1971

17 questions

1981

16 questions

1991

23 questions

2001

23 questions

2011

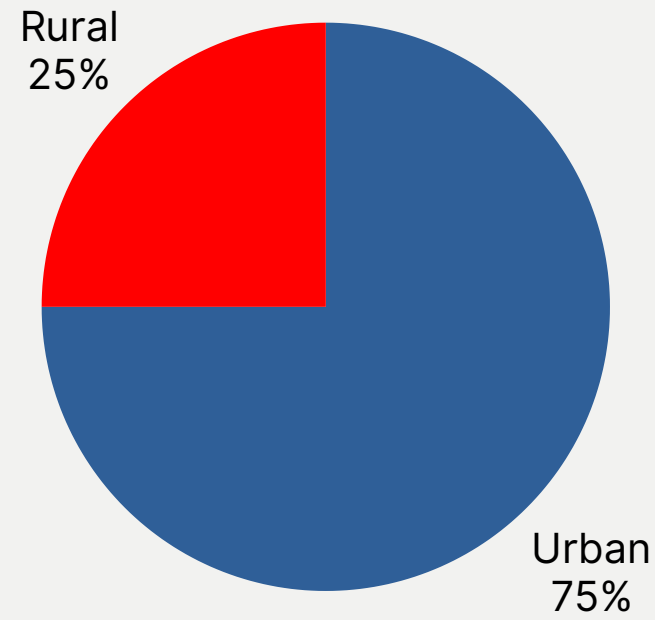
29 questions

History of changing trends of questionnaire

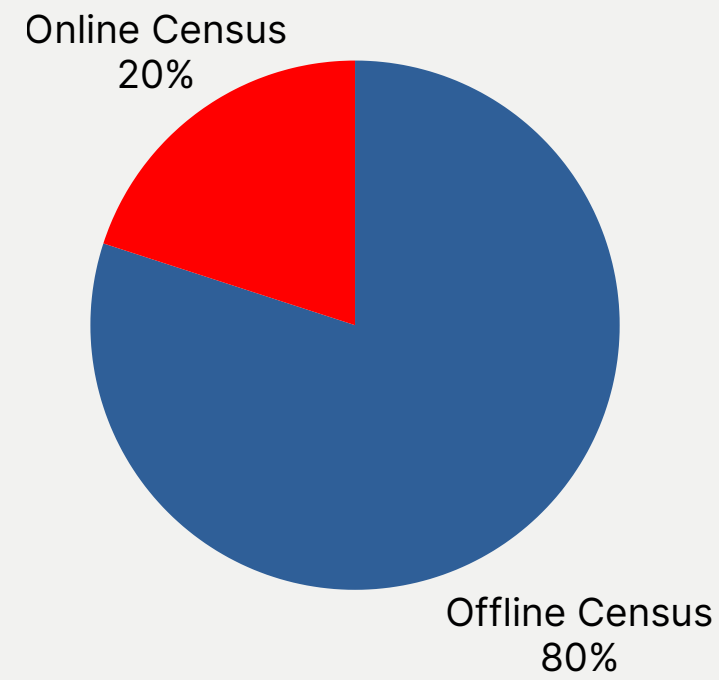
Questions of survey we performed:

- 1) Urban/Rural
- 2) What do you prefer?
online census , offline census
- 3) Place of origin ?
- 4) Current residence
- 5) Problem faced during offline census ?
- 6) What is your preferred language ?
Hindi ,English ,Other
- 7) Level of digital literacy
- 8) Employment status
- 9)What government IDs do you have
- 10) Do you feel social included in your region?
- 11) Do you look forward to move abroad ?

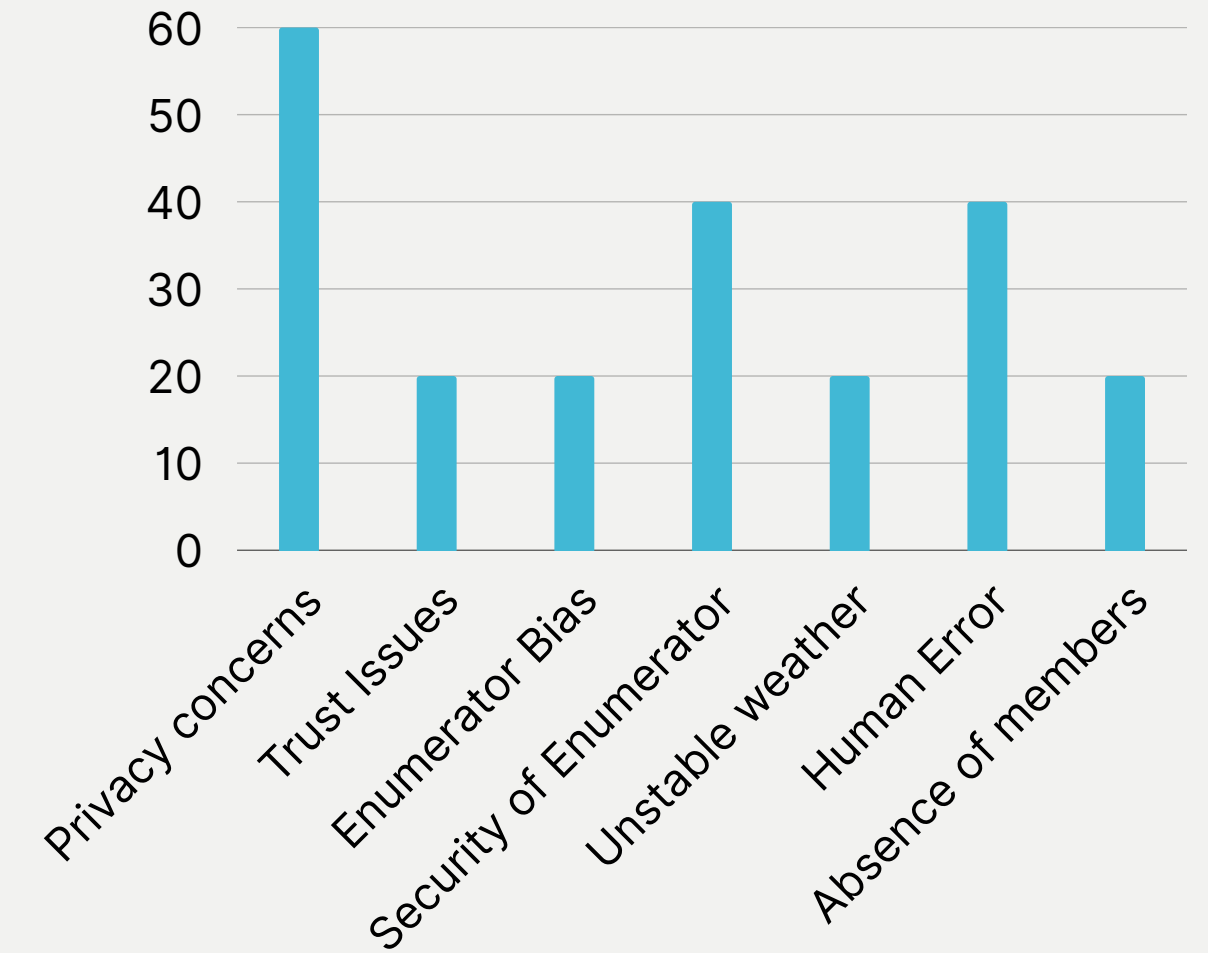
urban/rural



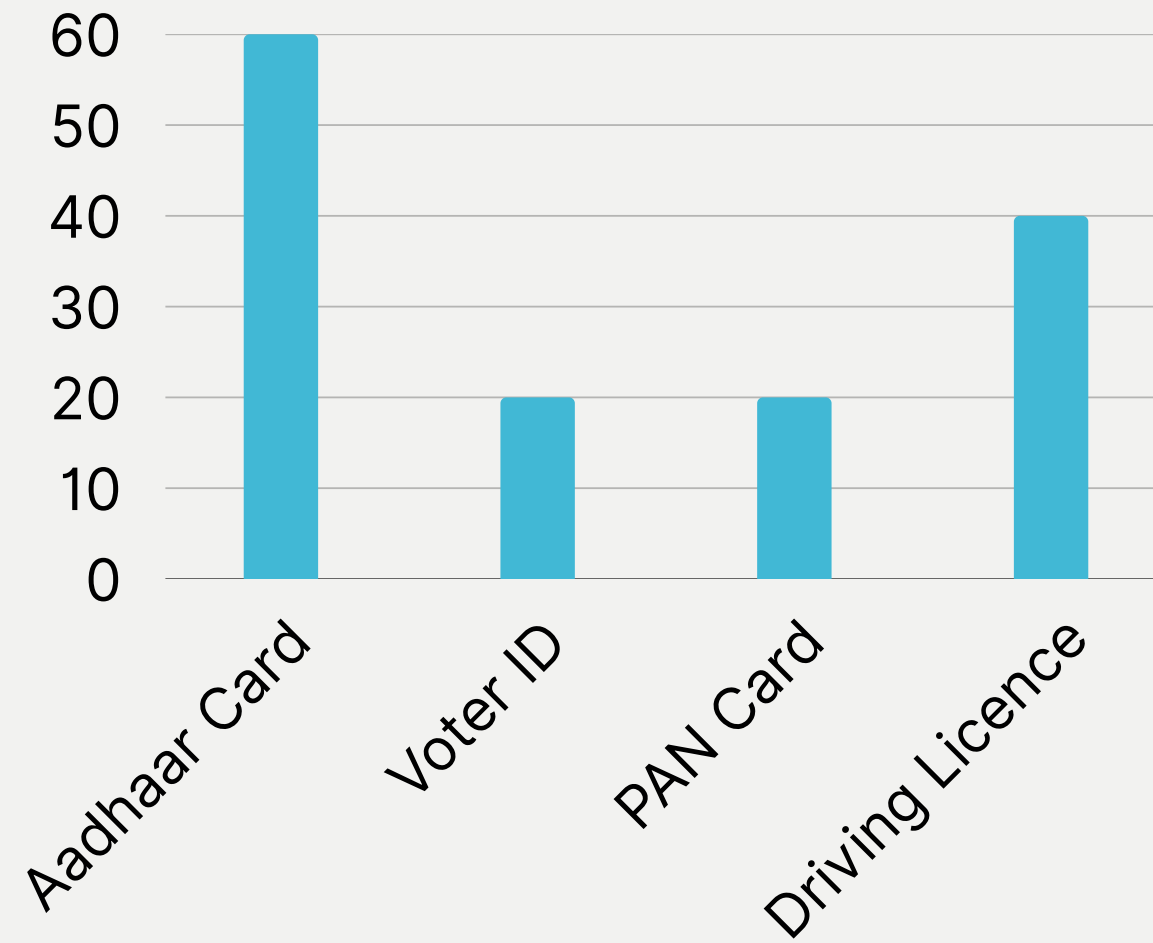
What do you prefer?



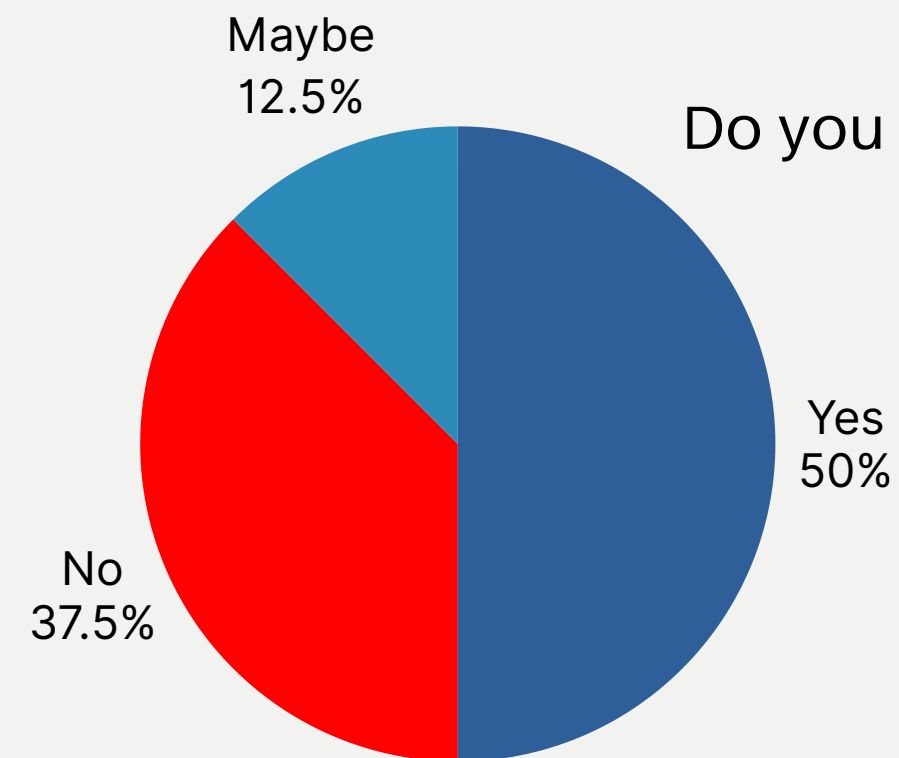
Problem faced during offline census



What government IDs do you have?



Do you look forward to move abroad ?





NAME: Akshay

AGE: 35

LOCATION : An urban construction site

EDUCATION : Some secondary education

OCCUPATION : Construction worker

MIGRATION STATUS : Permanent interstate migrant

CHARACTERISTICS:

Basic digital literacy

Smartphone use

CHALLENGES AND NEEDS:

- Needs simple, fast, no-paper process
- Wants clarity of documents required
- Often misses enumerator visits due to work timings

User Persona



NAME: Sushila

AGE:62

LOCATION : Village

EDUCATION : Limited formal education

OCCUPATION : Retired

DOCUMENT AVAILABILITY :

Possesses old documentation, like voter ID

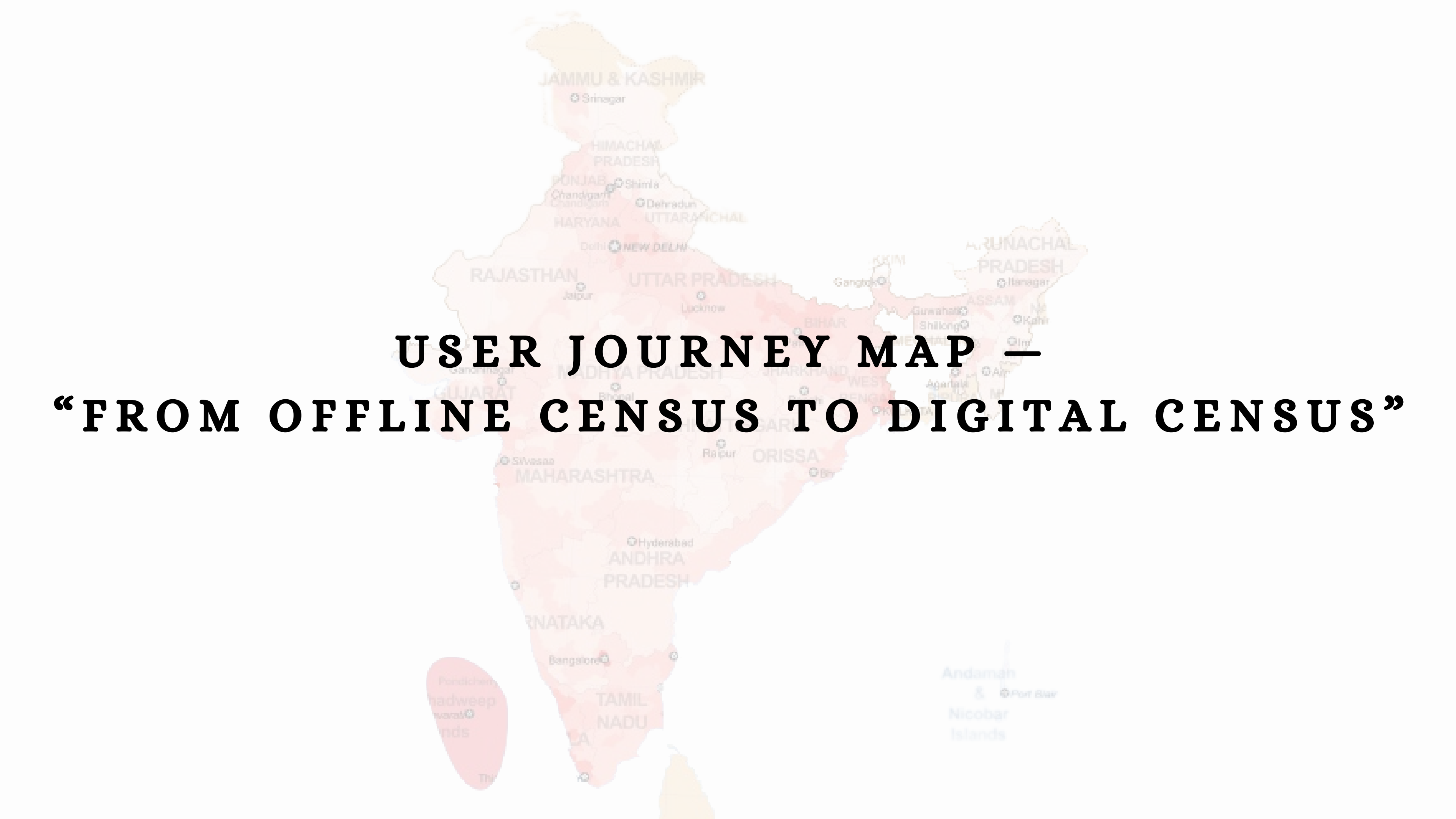
CHARACTERISTICS:

Low digital literacy

No internet access

CHALLENGES AND NEEDS:

- Needs assistance for form completion
- Prefers local language
- Requires trust + reassurance for privacy



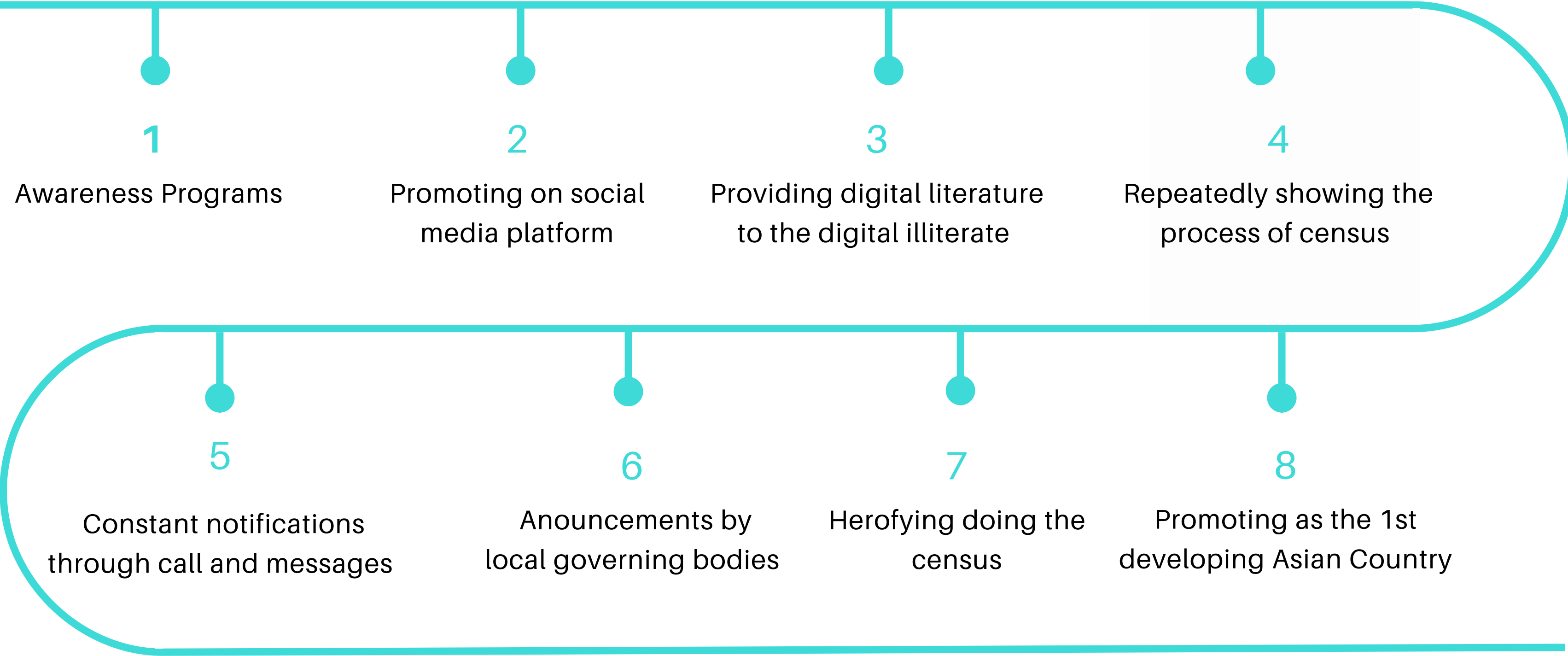
USER JOURNEY MAP — “FROM OFFLINE CENSUS TO DIGITAL CENSUS”

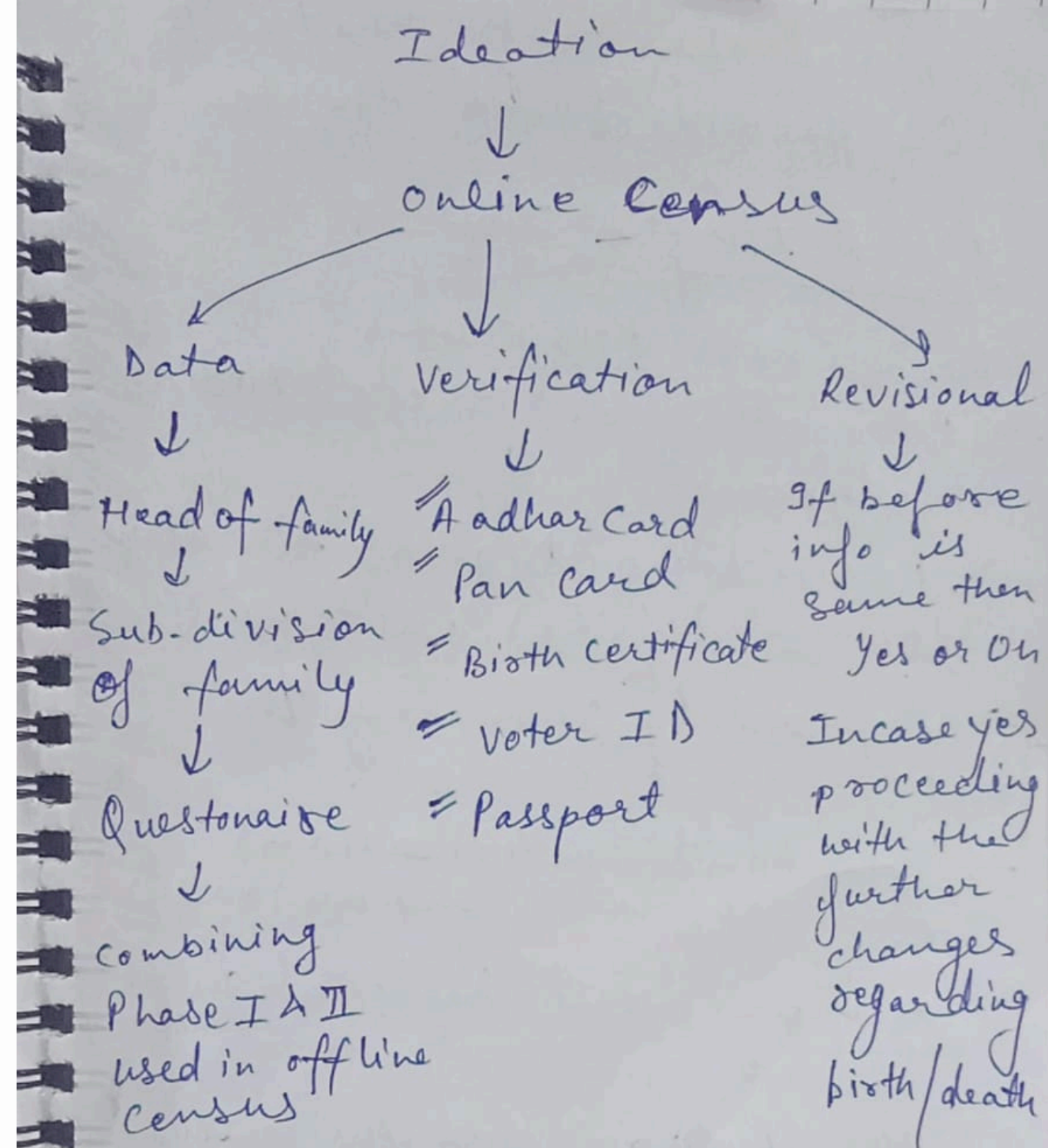
User: Akshay (Urban construction worker)				
Stage	What the User Does	What the User Feels	Pain Points	Opportunities
1. Awareness	Hears about census through gov announcements / social media.	Curious but unsure.	Unsure where to start, doesn't understand process.	Clear awareness campaign + simple instructions.
2. Accessing the System	Opens website/app to register.	Slightly nervous about digital steps.	Low digital confidence, language barriers.	Multi-language support, simple UI, guided onboarding.
3. Filling the Form	Enters personal + family details.	Wants to finish quickly.	Fear of making mistakes, confusion about required documents.	Smart input validation, real-time help, document checklist.
4. Verification	Uploads ID / completes OTP verification.	Relieved when it works smoothly.	Bad internet, unclear instructions, failed verification.	Offline mode + retry options + clear error messages.
5. Submission	Reviews and submits form.	Feels accomplished.	Worries if data was saved correctly.	Confirmation message + downloadable receipt.
6. After Submission	Waits for approval / confirmation.	Wants reassurance.	No updates, doesn't know status.	Status tracking + SMS/WhatsApp updates.

User: Sushila(62-year-old, rural resident)				
Stage	What She Does	What She Feels	Pain Points	Opportunities
1. Awareness	Hears from local govt worker.	Confused but willing.	Doesn’t understand digital census.	Door-to-door awareness + simple brochures.
2. Getting Help	Asks family member for assistance.	Relieved but dependent.	No digital literacy, no smartphone.	Assisted digital booths + village centres.
3. Filling Details	Provides answers to helper.	Wants it to be over soon.	Long form, unclear questions.	Simplified forms + audio instructions.
4. Verification	Shares existing ID details.	Worried about privacy.	Distrust in sharing information digitally.	Build trust with “your data is safe” messaging.
5. Submission	Completes with help.	Relieved.	Unsure if submitted properly.	Printed receipt / SMS confirmation.
6. After Submission	Waits for confirmation.	Curious + hopeful.	No feedback or updates.	Simple “Your census is completed” message.

PUBLICITY

Timeline





Incase of weaker sections

→ Govt employees in regions all over india will be assigned incase someone is not able to do or does not have the tech. they can call and reach out

Ideation

Online census

1

2

3

4

5

Enter Aadhaar number

*Suggested Head-Of-Family

SEND OTP

Online census

1

2

3

4

5

Enter OTP

SUBMIT

Did not receive OTP?

RESEND OTP

Online census

1

2

3

4

5

Help ***** congratulations on becoming a hero by contributing for the cause of the nation. Please enter the following data with utmost honesty and accuracy.

Enter your building number

Enter census house number provided

Use of house

Condition of house

Household Phone Number

Total people living in a house

Number of rooms

Number of married couples

Source of drinking water

Availability of drinking water

Source of lighting

Washroom within premise

Type of washroom facility

SUBMIT

Online census

1

2

3

4

5

Waste water outlet connection

Kitchen availability

Fuel used for cooking

Bathing facility

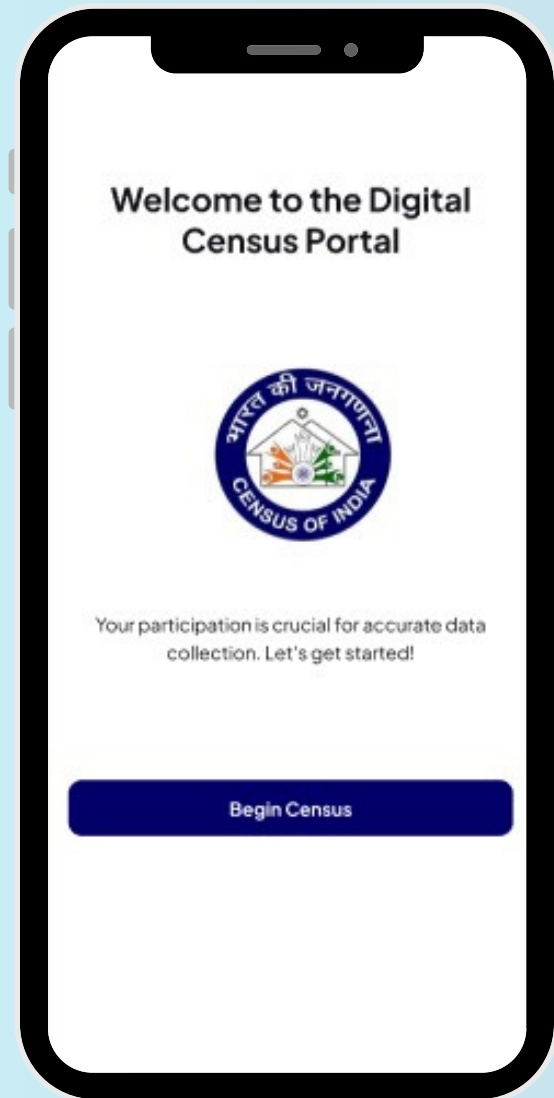
Television

Computer or Laptop

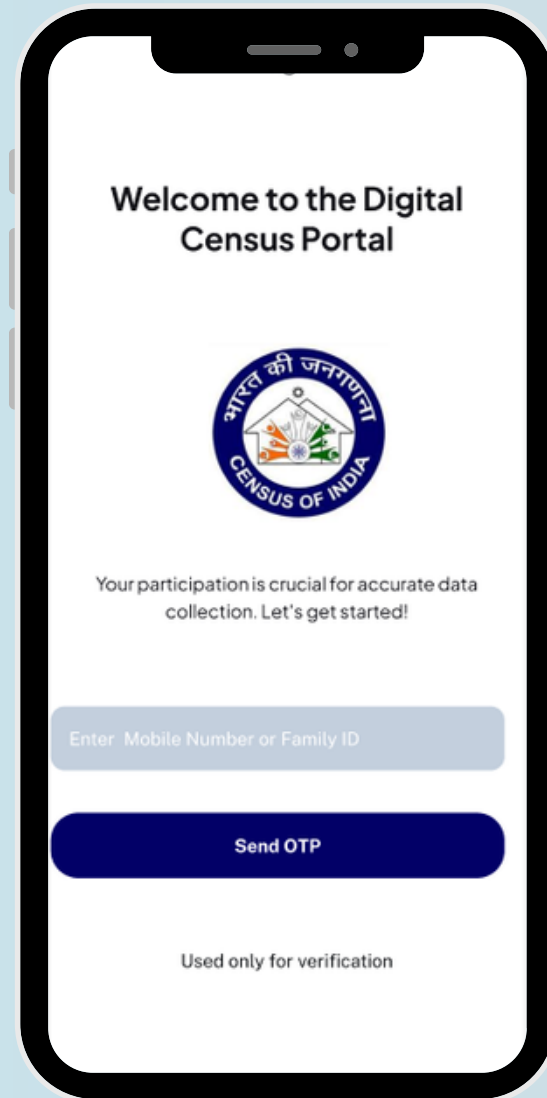
Number of mobile phones

SUBMIT

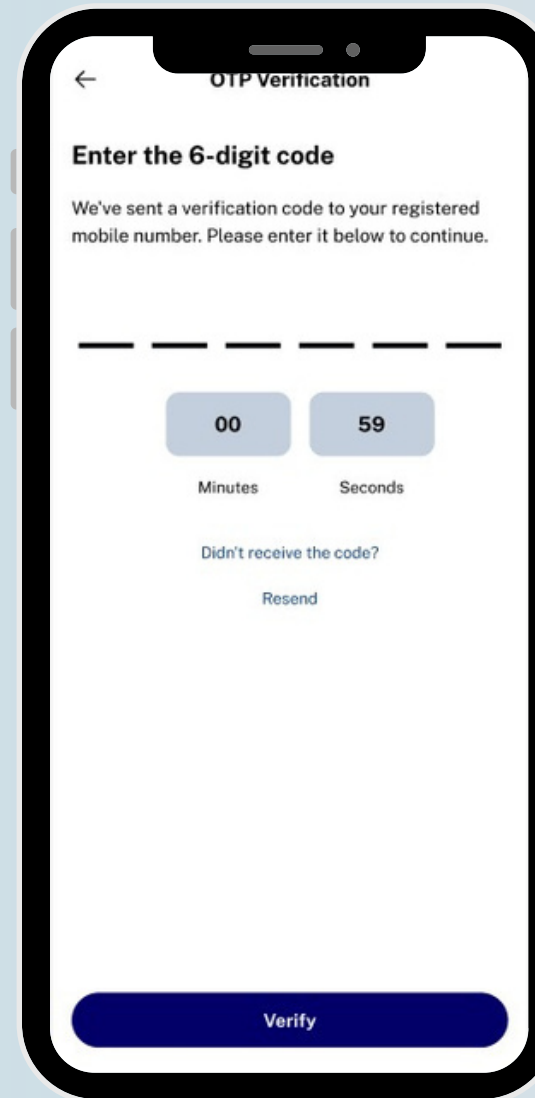
Prototype



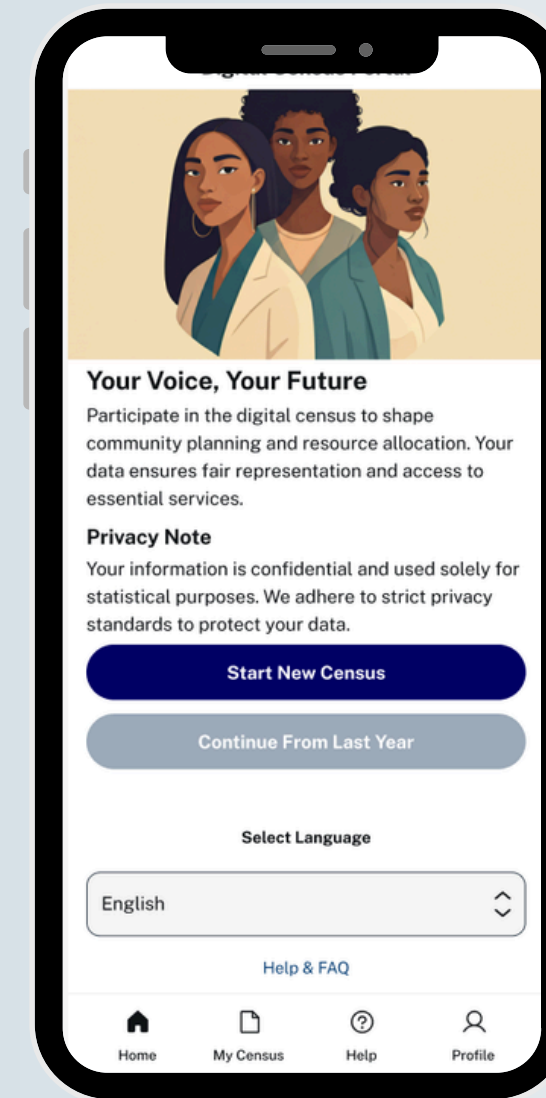
Welcome screen



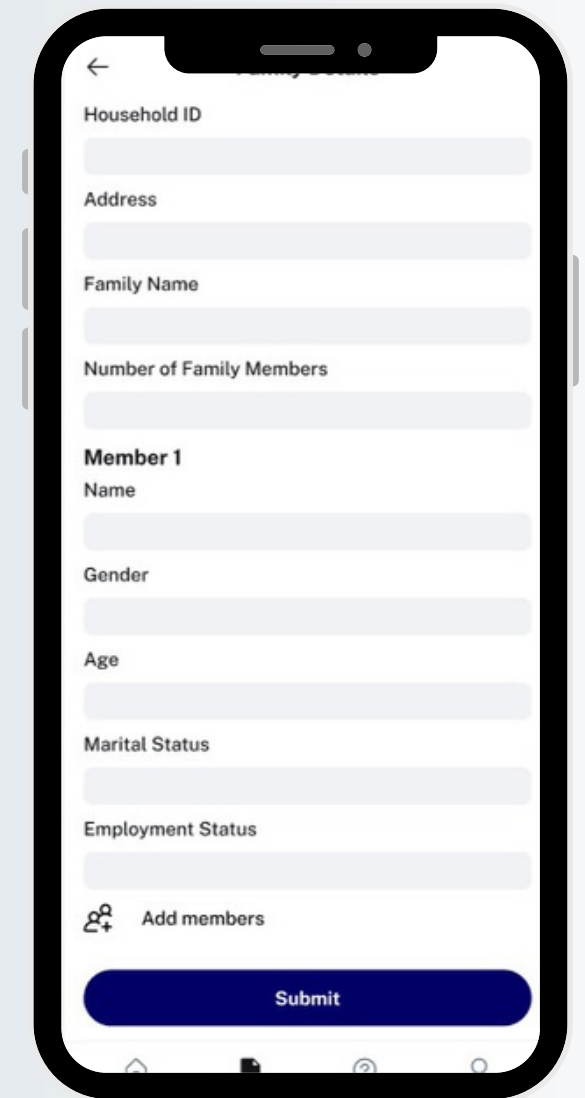
Login Page



OTP Verification

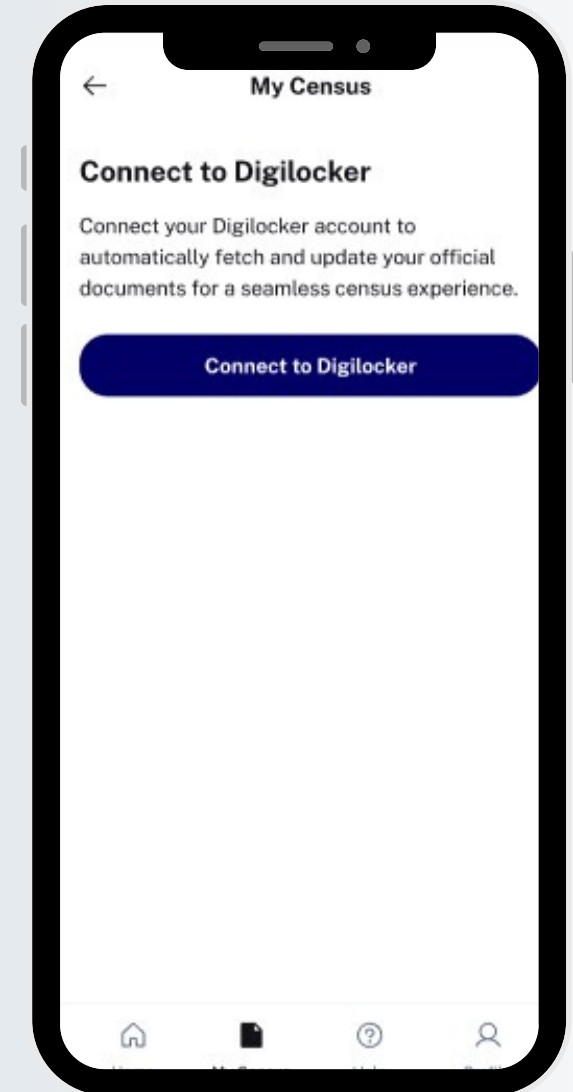
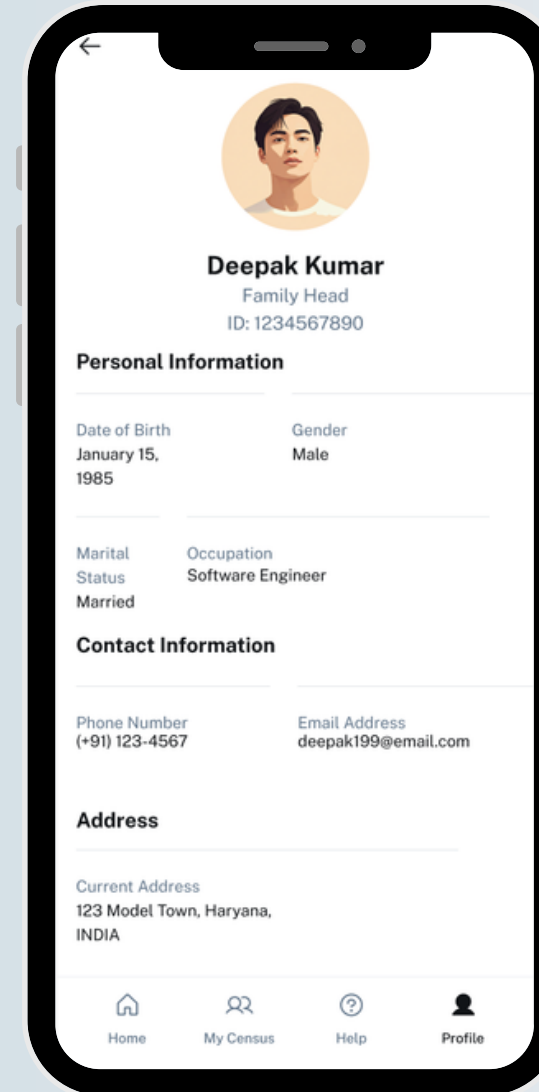
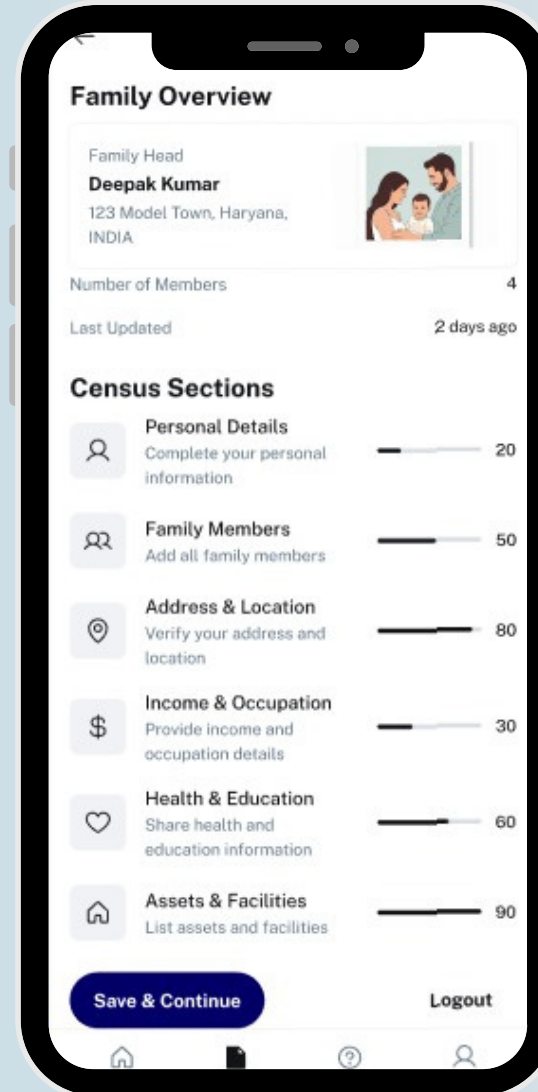
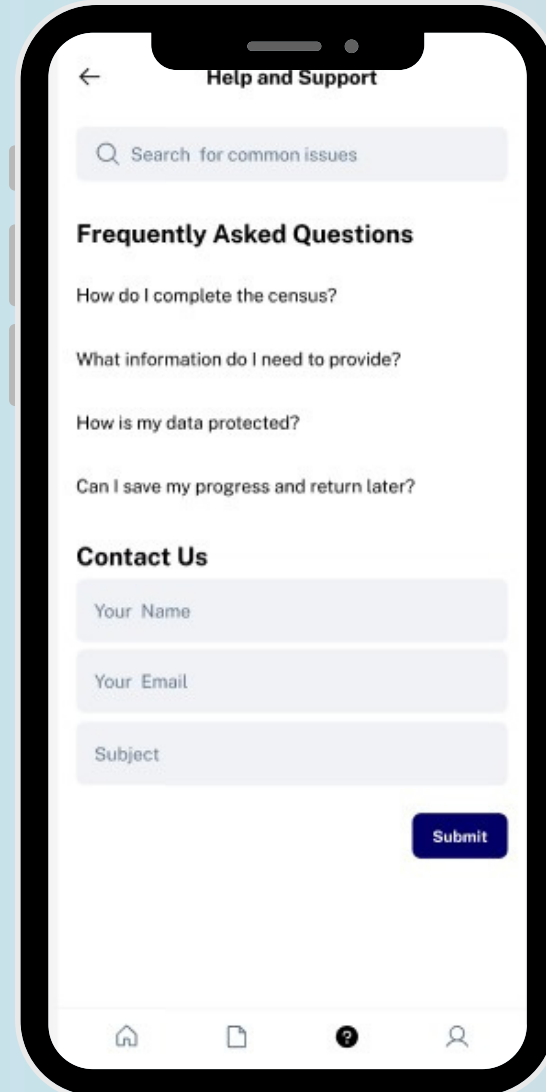
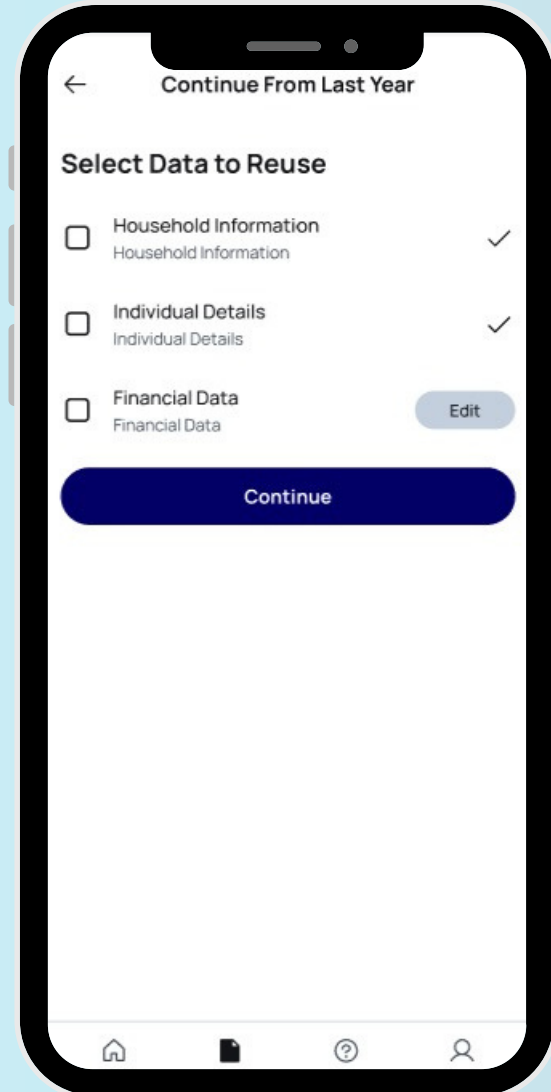


Home Page



New Census

Prototype



Thank You